



Synthetic chicken free eggs

San Francisco-based 'Every Company' releases its synthetic egg whites made without chickens. <https://digit.in/apr22-17>



Photos into 3D scenes

NVIDIA's tiny new AI transforms photos into full 3D scenes in mere seconds. <https://digit.in/apr22-18>

Intel Core i7-12700K Desktop Processor

Just as good as the 12900K

Intel's Alder Lake processors have pretty much given Intel the lead in the consumer CPU space with their new hybrid architecture working wonders in a lot of use cases. It did turn out that disabling certain cores within the Alder Lake flagships have certain benefits but the overall package with the efficiency cores still makes for a powerful combination. The Intel Core i7-12700K is more value for money among the flagships right off the bat with its much lower price tag but does the performance match up to its Core i9 sibling? That's what we're looking to find out.

SPECIFICATIONS

The Intel Core i7-12700K has the same amount of performance cores as the 12900K. The difference in core counts is purely because of the four fewer Efficient Cores. Since most of the performance is driven by the P-Cores, this would mean that the two processors should ideally be very close with just the clock speeds and available cache being the only factors leading to the relative performance shortfall. Speaking of clock speeds, the P-Cores are running at 3.6 GHz whereas the E-Cores are running at 2.7 GHz. At boost, the P-Cores hit 5 GHz and the E-Cores can go up to 3.8 GHz. The overall thread count is at 20 for the 12700K and 24 for the 12900K. Again, only a few heavily threaded applications will differ in performance going by all the differences in specifications.

PERFORMANCE

In the single-threaded Cinebench R23 benchmark, the Intel Core i7-12700K scores 1940 points which when compared to its predecessor, the Intel Core i7-11700K makes for a difference of 340 points. The AMD Ryzen 9 5950X



is at 1640 points and the 5900X is at 1622 points.

Even in the multi-threaded run, the Intel Core i7-12700K manages to score 22230 points which isn't that far off from the 12900K that had scored 27770 points. The AMD Ryzen 9 5950X scores more than the 12700K with 25815 points. The 5950X has 32 threads whereas the 12700K has 20 threads. The per-thread score is still higher with the Intel processors.

In Blender, the Core i7-12700K finished the entire run in about 115 seconds which puts it ahead of the AMD Ryzen 5 5900X but behind the 5950X. Content creation software such as V-Ray and Keyshot also put the AMD Ryzen 9 5900X ahead of the 12700K.

Coming to productivity tasks, the Core i7-12700K comes very close to the 12900K in most benchmarks comprising of software such as Microsoft Word, Excel, Powerpoint, etc. In WinRAR, the Core i7-12700K is ahead of the Ryzen 9 5900X but behind the 5950X by a smidge. Switching to 7-Zip puts both AMD flagships ahead of the Core i7-12700K. Even the Ryzen 7 5800X

goes neck-and-neck during the decompression benchmark.

GAMING (DISCRETE GPU)

In our gaming benchmarks, we run several new titles on the NVIDIA RTX 2080 Ti. There are a host of different games based on a variety of current-gen game engines using different graphics APIs such as DirectX11, 12, Vulkan and OpenGL.

When gaming at 1440p resolution, we see that the Intel Core i7-12700K is practically the same as the 12900K. In most games, the two processors only differ by less than one per cent at most when comparing FPS numbers. This includes titles such as Shadow of the Tomb Raider, The Witcher 3, Assassin's Creed Odyssey and Strange Brigade. The story is the same when gaming at 1080p as well. The AMD Ryzen 9 5900X and the Ryzen 7 5800X are

next and neck with the 12700K in quite a few titles. So you've got a healthy set of competing processors to choose from for your next gaming PC build. It all comes down to how much the overall configurations cost.



PERFORMANCE.....87
VALUE FOR MONEY...75

VERDICT

There aren't that many applications that the average joe will be using that can benefit from the additional four threads that the 12900K sports. Considering that there is a price difference of 15K between the Alder Lake flagship and the 12700K, one would assume that the performance might be worlds apart. From our benchmarks, it seems that the Intel Core i7-12700K is practically the same as the 12900K making it a much better processor for your next gaming PC.

—Mithun Mohandas

SPECIFICATIONS

CORES: 12 | THREADS: 20 | BASE CLOCK (P/E): (3.6 / 2.7) GHz | BOOST CLOCK (P/E): (5.0 / 3.8) GHz | PROCESSOR NODE: Intel 7 | L3 CACHE: 25 MB | L2 CACHE: 1.25 MB per core | MAX TURBO POWER: 190 W.

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